CM5 NANO COMPATIBLE — FUNCTIONAL CARRIER — ENGINEERING DESIGN

55 × 40 mm target | CM5 / CM5 Lite | USB-C 5 V input + USB0 recovery | single USB3 host | dual 4-lane MIPI | GPIO | SD | fan | RTC

New physical parts and reviewed contact mappings. Prototype verification and PCB layout remain required. Do not order from schematic alone.

CM5 module contacts / GPIO voltage selection / controls

H2: fit shunt 2–3 for default 3.3V; 1–2 selects 1.8V. Power OFF before changing.
GPIO bank follows VREF; header 3.3V pins remain 3.3V.
MIPI1 / HAT I2C follows GPIO_VREF; verify peripherals in 1.8V mode.

USB-C power + recovery

BOOT: hold during power-up for USB recovery.
POWER: module power button. CM5 eMMC uses onboard storage; external microSD applies only to CM5 Lite.

USB3 SuperSpeed ESD

CM5 Lite microSD supply

3.3 V peripheral regulator — sequenced by CM5_3V3

Host power and indicators

Fan and RTC

MIPI0 protection

MIPI1 protection

USB-C is the 5V input and USB2 device/recovery port.
CM5 CC1/CC2 negotiate power; no external 5.1k Rd.
GPIO 5V is directly the same rail; use one supply at a time.

Power declarations: USB-C supplies 5V and GND;
GPIO_VREF is supplied through the fitted H2 shunt (default 2–3).

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